

Exterior and Interior Angles of Polygons

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An **interior angle** is an angle formed inside a polygon at each vertex, between two adjacent sides.

An **exterior angle** is an angle formed by one side of a polygon and the extension of the adjacent side at a vertex.

Linear Pair

The **exterior angle** and its **adjacent interior angle** form a **linear pair**.

$$\text{Exterior Angle} + \text{Adjacent Interior Angle} = 180^\circ$$

Finding the Exterior Angle

To find the exterior angle, subtract the adjacent interior angle from 180° .

$$\text{Exterior Angle} = 180^\circ - \text{Adjacent Interior Angle}$$

Finding the Interior Angle

To find the adjacent interior angle, subtract the exterior angle from 180° .

$$\text{Adjacent Interior Angle} = 180^\circ - \text{Exterior Angle}$$

Examples

Example 1

An interior angle of a polygon measures 115° .
Find its adjacent exterior angle.

$$180^\circ - 115^\circ = 65^\circ$$

The exterior angle is 65° .

Example 2

An exterior angle of a polygon measures 42° .
Find its adjacent interior angle.

$$180^\circ - 42^\circ = 138^\circ$$

The interior angle is 138° .

Example 3

An interior angle of a polygon is three times its exterior angle. Find both angles.

$$x + 3x = 180^\circ$$

$$4x = 180^\circ$$

$$x = 45^\circ$$

Exterior: 45° Interior: 135°

Practice Exercises

Practice 1

The interior angle of a regular hexagon is 120° .
Find each exterior angle.

Practice 2

At a vertex of a polygon, the interior angle measures 135° . Find the adjacent exterior angle.

Practice 3

At a vertex of a polygon, the exterior angle measures 68° . Find the adjacent interior angle.

Practice 4

The interior angle of a regular octagon is 135° .
Find each exterior angle.

Practice 5

The exterior angle of a regular polygon is 45° .
Find the interior angle.

Practice 6

The interior angle of a polygon is twice its exterior angle. Find both angles.

Practice 7

The exterior angle of a regular polygon is 30° .
Find the interior angle.

Activities

Activity 1

An interior angle of a polygon measures 72° .
Find the measure of its adjacent exterior angle.

Activity 2

An exterior angle of a polygon measures 63° .
Find its adjacent interior angle.

Activity 3

The interior angle of a regular octagon is 135° .
Find each exterior angle.

Activity 4

At a vertex of a polygon, the interior angle measures 128° . Find the adjacent exterior angle.

Activity 5

At a vertex of a polygon, the exterior angle measures 55° . Find the adjacent interior angle.

Activity 6

The interior angle of a regular decagon is 144° .
Find the measure of each exterior angle.

Activity 7

At a vertex of a polygon, the interior angle measures 150° . Find the adjacent exterior angle.

Activity 8

At a vertex of a polygon, the exterior angle measures 38° . Find the adjacent interior angle.

Activity 9

The interior angle of a regular hexagon is 120° .
Find the measure of each exterior angle.

Activity 10

The interior angle of a regular pentagon is 108° . Find the measure of each exterior angle.

Thank you!